gravitational pull on the other galaxies. The galaxies' motions were so fast that there was no way their gravitational pulls could hold the cluster together. Our best understanding of the universe and of gravity insisted that the cluster must be flying apart, and would soon be completely destroyed. If so, then the cluster must have formed by random motions of all those galaxies and would disrupt in a veritable blink of an eye compared to other astronomical phenomena.

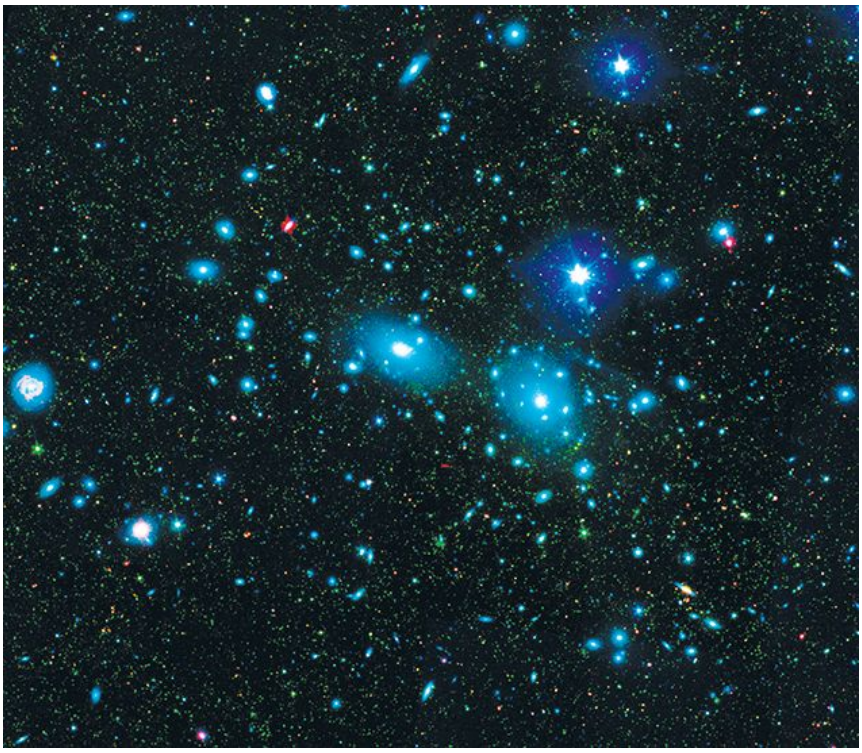


Fig. 24.2. The Coma cluster of galaxies as seen through a large telescope.

This conclusion was totally implausible to Zwicky. Something was wrong with our conventional wisdom. Zwicky made an educated guess: The Coma cluster must be filled with some sort of "dark matter" whose gravity is strong enough to hold the cluster together.

Now, many anomalies that astronomers and physicists think they have discovered go away when observations improve. This one did not.